

Nicole F Dear

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EDUCATION

Michigan State University

PhD, Geography, Environment and Spatial Sciences

East Lansing, MI

Expected 2026

University of Michigan

MPH, Global Health Epidemiology

BS, Microbiology

Ann Arbor, MI

May 2016

May 2014

RESEARCH EXPERIENCE

Henry Jackson Foundation (in support of the Military HIV Research Program)

Bethesda, MD/Remote

Biostatistician III

April 2022 – August 2022; January 2023 – December 2024

- Project oversight manager for an ongoing longitudinal cohort study in four African countries and a multi-country emerging infectious disease surveillance study
- Conducted statistical analysis of data from cross-sectional studies, longitudinal cohort studies and clinical trials and communicated findings to scientific and non-technical audiences via written and oral modalities
- Co-authored 8 peer-reviewed publications

Biostatistician II

July 2019 – March 2022

- Conducted analysis of data from two cross-sectional pediatric HIV studies in Kenya and Tanzania, a prospective HIV incidence study in Thailand and a US-based cohort study to assess STI reinfection in a military population
- Conducted statistical analyses for a longitudinal cohort study in four African countries to evaluate impact of clinical practices and biological and socio-behavioral factors on HIV infection and disease progression
- Generated, interpreted and communicated findings to scientific and non-technical audiences and provided data-driven evidence for PEPFAR programming, HIV clinical care and future vaccine studies
- Coordinated with regulatory, laboratory, data and clinical teams to develop and implement protocol amendments, survey questions and logic checks to ensure data quality and consistency
- Provided epidemiologic, biostatistical and scientific writing support to internal, external and in-country staff
- Co-authored 17 peer-reviewed publications

Peace Corps Response

Durban, South Africa

Monitoring and Evaluation Specialist

November 2018 – June 2019

- Developed an online data collection instrument allowing interviewers to directly capture participant survey data
- Reduced data errors by 60% by creating an automated tool in R to identify patterns of missing, out of range and inconsistent data in a multi-site study, highlighting areas for data entry staff retraining
- Analyzed clinical and program data and delivered written program evaluation report to funders
- Trained 15 staff members in Microsoft Excel over a six-month period

Colorado Haiti Project

Port-Au-Prince, Haiti/Remote

GIS Consultant

August 2018 – June 2020

- Built organizational capacity for spatial data capture, management and visualization
- Designed online interactive map for client facing website

Centers for Disease Control

Port-Au-Prince, Haiti

PHI/CDC Global Health Surveillance Fellow

September 2016 – August 2018

- Increased local capacity by training 25 in-country staff in GIS software and methods leading to identification of resource gaps/overlaps and areas of high disease burden
- Mapped new cases of disease, validated spatial data and maintained online outbreak response dashboards for the Global Polio Eradication Initiative
- Provided technical support to the Ministry of Health to improve timely use of national and program data and make informed decisions about the national HIV strategy

- Coordinated logistics for Haiti's first Population-based HIV Impact Assessment
- Identified target areas for resource allocation after Hurricane Matthew by adapting a data collection tool in a post-disaster setting, analyzing and summarizing evidence-based findings

JEN Associates, Inc.

Junior Health Data Analyst

Ann Arbor, MI

April – August 2016

- Provided technical assistance to government agencies and non-governmental organizations for patient consent, data collection, analyses and risk profiling of patients from a 5% sample of Medicare and Medicaid records

University of Michigan

Research Assistant

Ann Arbor, MI

September 2014 – April 2016

- Extracted satellite derived variables and analyzed them in combination with household survey data to identify factors associated with increased mosquito abundance and patterns of vector distribution in Blantyre, Malawi
- Fostered collaborative working relationships across four institutes in the United States and Malawi
- Conducted a systematic review to inform future research directions

SELECT PUBLICATIONS

Esber A, **Dear N**, King D, Francisco L, Sing'oei V, et al. Achieving the third 95 in sub-Saharan Africa: application of machine learning approaches to predict viral failure. *AIDS* 37(12):p 1861-1870. 2023. doi: 10.1097/QAD.0000000000003646

Dear N, Esber A, Iroezindu M, Bahemana E, Kibuuka H, et al. Routine HIV clinic visit adherence in the African Cohort Study. *AIDS Res Ther.* 2022;19(1):1. doi: 10.1186/s12981-021-00425-0. PMID: 34996470; PMCID: PMC8742415.

Dear N, Francisco L, Pitisutthithum P, Nitayaphan S, Schuetz A, et al. Unique HIV Risk Factors and Prevention Needs for Transgender Women and Cisgender Men Who Have Sex with Men in Bangkok, Thailand. *Transgender Health.* 2022.

Complete list: <https://scholar.google.com/citations?user=Dj4m8eAAAAAJ&hl=en&oi=ao>

CONFERENCE PRESENTATIONS

Dear N and Grady SC. COVID-19, place-based vulnerabilities, and maternal and infant health in Michigan. Oral Presentation. International Medical Geography Symposium (IMGS). July 2024.

Dear N, Thotakura B, Grady SC. Racial Residential Segregation and Sexually Transmitted Infections during Pregnancy in the Detroit Metropolitan Area. Poster. GEOMED, Virtual. October 2022.

Dear N, Grady SC, Rivera A. Advancing Research on Racial Residential Segregation for Health Studies. East Lakes Division/West Lakes Division American Association of Geographers Joint Virtual Annual Meeting. October 2021.

Dear N, Esber A, Parikh A, et al. Impact of COVID-19 among people living with HIV in the African Cohort Study. Poster. CROI, Virtual. March 2021.

Dear N, Esber A, Iroezindu M, et al. Factors Associated with Transactional Sex in the African Cohort Study. Poster. HIV Research for Prevention, Virtual. January 2021.

Dear N, Esber A, Iroezindu M, et al. Routine HIV Clinic Appointment Adherence in the African Cohort Study. Poster. IAS, Virtual. July 2020.

Dear N, Kadangwe C, Mzilahowa T, et al. Spatial Patterning and Fine-Scale Heterogeneity of Malaria Risk Along an Urban-Rural Continuum in Blantyre, Malawi. Poster. ASTMH, Baltimore, MD. November 2017.

TEACHING EXPERIENCE

Michigan State University

East Lansing, MI

onGEO Professional Certificate: GEO113 (Economic Geography) (Online), Instructor

Summer (Session 2) 2025

onGEO Professional Certificate: GIS325 (Geographic Information Systems); IGT325 (Introduction to Geospatial Technologies) (Online), Course Grader

Summer (Session 1) 2025

GEO 424: Advanced Remote Sensing (In-person), Lab Instructor

Spring 2025

onGEO Professional Certificate: GIS524 (Geographic Information Systems); IGT524 (Introduction to Geospatial Technologies); CART624 (Cartography) (Online), Course Grader

Fall 2024

ISS 310: People and Environment (Online), Instructor

Fall 2023; Spring 2024

GEO 325: Geographic Information Systems (Online), Instructor

Spring 2023

GEO 866: Spatial Data Analysis (In-person), Lab Instructor

Fall 2022